CLASS 10 · CBSE · NCERT MATHEMATICS · FORMULA BANK

# Chapter 7: Coordinate Geometry

Every formula and result of the chapter on one sheet

| | | |
|---|---|---|
| 1 | Distance between two points | $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ |
| 2 | Distance from the origin | $\sqrt{x^2 + y^2}$ |
| 3 | Section formula (internal, $m_1 : m_2$) | $\left(\frac{m_1 x_2 + m_2 x_1}{m_1 + m_2}, \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2}\right)$ |
| 4 | Section formula (ratio $k : 1$) | $\left(\frac{k x_2 + x_1}{k + 1}, \frac{k y_2 + y_1}{k + 1}\right)$ |
| 5 | Mid-point | $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$ |
| 6 | Centroid of a triangle | $\left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}\right)$ |
| 7 | Point on the x-axis | $(x, 0)$ |
| 8 | Point on the y-axis | $(0, y)$ |
| 9 | Collinear points | $AB + BC = AC$ for some order of the three points |
| 10 | Parallelogram | Diagonals have the same mid-point |
| 11 | Points of trisection | Divide the segment in the ratios $1 : 2$ and $2 : 1$ |